



The Journey Of TV Technology In India

Today, public demand is towards video information and entertainment, and preferred choice of the user terminal is the ever-present smartphone



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Television (TV) technology has undergone substantial changes since its arrival in India six decades ago. The first TV transmitter was set up in 1959 at New Delhi, followed by Mumbai in 1972. Further progress in TV broadcasting was made in 1976, when India embarked on Satellite Instructional TV Experiment (SITE) via ATS 6 satellite taken on loan from NASA, and educational TV programmes were beamed to India from ATS 6. Users were introduced to dish antenna as a means to receive TV programmes.

Terrestrial transmitters were also started by Doordarshan India (DDI) in VHF band I (41MHz - 68MHz) and band III (174MHz - 216MHz). In the initial phase, carrier frequencies in the UHF and C bands were used for satellite broadcasting, which required a large dish antenna for reception. Satellite TV gained popularity when Ka and Ku band receivers were developed, and a small 60cm dish antenna could be used to receive signals from geostationary satellites.

Antenna on the geostationary satellites can be designed to generate a footprint required to cover the target geographical area. Thus, tailor-made beams can be gen-

erated to save power as well as take care of possible objections from population outside the boundary.

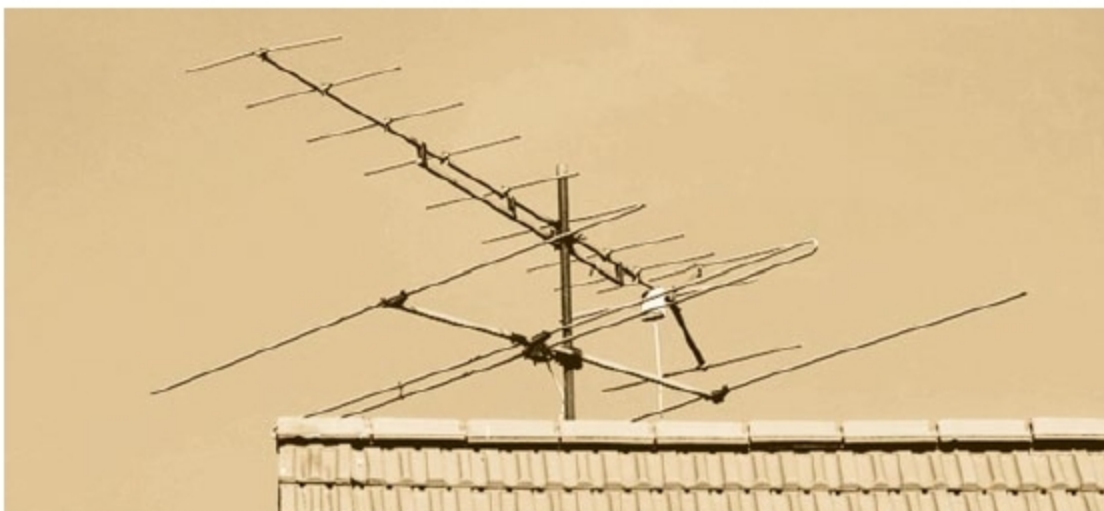
Developments in the field of data compression have brought dramatic reduction in bandwidth requirements. Encryption of data bit-streams also facilitates control of paid channels by service providers. Paid channels have gained consumer confidence, and the concept of free-to-air (FTA) channels has faded with time. The analogue TV finally closed down in India on January 13, 2018. As most of the viewers had already switched over to satellite-based DTH systems, no one even noticed it!

Analogue TV

The analogue TV, operating in VHF channels on standard PAL B and G bands, transmits video signal modulated on the carrier using vestigial side-band (VSB) technique. Accompanying sound frequency is modulated on a separate carrier frequency. Total bandwidth of the TV signal is about 7MHz or 8MHz.

VSB signal is decoded in the receiver, and luminance as well as colour information is separated. The audio carrier is demodulated to recover audio channel. These three output signals, composite video containing luminance and colour signals, along with left and right stereo audio channels, together constitute the intelligence to be presented on the display device and speakers.

SITE TV programmes were beamed to ground at 860MHz, using frequency modulation for video signals. Indian Space Research Organization (ISRO) took on a major role and undertook the responsibility of designing high-reliability receiver sets. ISRO established regional centres for generation of programmes, handling of community TV sets and uploading of programmes to the satellite.



Terrestrial analogue TV Yagi antenna (Credit: www.datareign.com)